JEE-Main-05-04-2024 (Memory Based)
[MORNING SHIFT]

Physics

Question: The given figure shows a system 3 blocks and a pulley. The pulley is light and smooth the strings are inextensible and light. If the right end of string connected 6 kg block is given fixed velocity of 3m/s in downward direction then find T_1(in N)

(Take $g = 10m/s^2$)

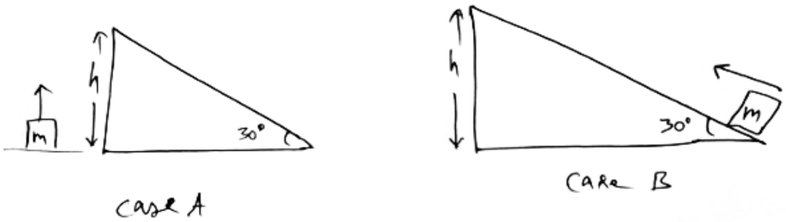
Options:

(a) 60
(b) 200
(c) 100
(d) 300
Answer: (b)

Question: In a YDSE setup the values of D, d and λ are 2m, 3 mm and 5000 Å respectively. Find the position of 3^{rd} bright fringe in mm
Options:
(a) 1 mm
(b) 5 mm
(c) 20 mm
(d) 15 mm
Answer: (a)

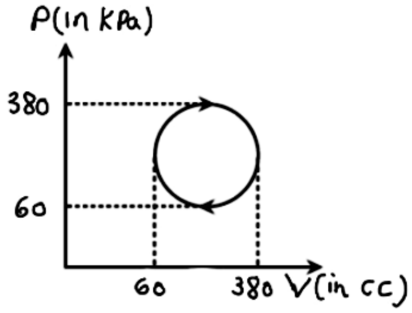
Question: Potential difference between the plates of capacitor of capacitance 12 μF changes by 40V at Frequency of 40 KHZ. Find displacement current
Options:
(a) 10 A
(b) 9.2 A
(c) 19.2 A
(d) 22.2 A
Answer: (c)

Question: A block of mass 'm' is raised by height, h = 20m in two ways as shown in diagram. In case A, block is raised vertically while in case B, block is raised along the incline. Then the ratio of work done by gravity in both cases is



Options:
(a) 1 : 2
(b) 2 : 1
(c) 1 : 1
(d) 1 : √2
Answer: (c)

Question: A simple pendulum has time period T_1 at a height R from earth's surface. If height of pendulum becomes 2R, time period is found to be T_2. Identify the correct relation
Options:
(a) $2T_1 = 3T_2$
(b) $4T_1 = 9T_2$
(c) $9T_1 = 4T_2$
(d) $3T_1 = 2T_2$
Answer: (d)

Question: The cyclic process forms a circle on a PV diagram as shown in the figure. The work done by the gas is (in joules):


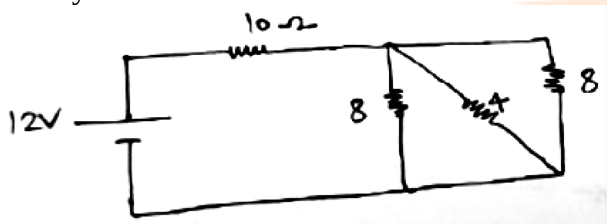
Options:
(a) -80.4
(b) +80.4
(c) 40.2
(d) -40.2
Answer: (b)

Question: If the energy density is represented by is u and gravitational constant by G, then which of the following quantity will have same unit as that of $\sqrt{uG}$
Options:
(a) $\frac{Force}{mass}$
(b) $\left(\frac{Force}{mass}\right)^2$
(c) velocity
(d) $(velocity)^2$

Answer: (a)

Question: Inner and outer wires of a coaxial wire carry equal and opposite current magnetic field is zero
Options:
(a) Outside both the wires
(b) Inside both the wires
(c) Between the wires
(d) No where
Answer: (a)

Question: Find the equivalent resistance of the following circuit and the current through the battery



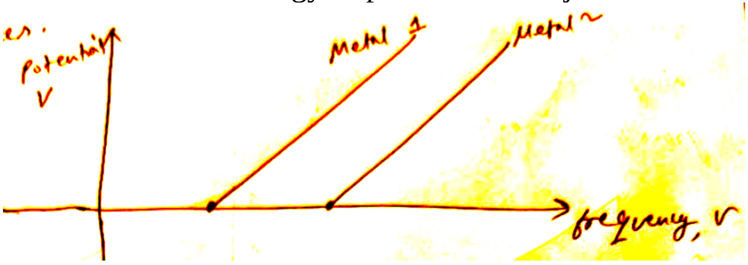
Options:
(a) 12Ω, 1 Amp
(b) 30Ω, 4 Amp
(c) 6Ω, 2 Amp
(d) 10Ω, 1.2 Amp
Answer: (a)

Question: Statement-1: It a capillary tube is dipped in a liquid, there is no fall (or) rise of liquid then angle of contact is 0°
Statement-2: Angle of contact depends on nature of capillary tube and liquid.
Options:
(a) Only statement-1 is correct
(b) Only statement-2 is correct
(c) Both are correct
(d) Both are incorrect
Answer: (b)

Question: Find the potential difference across capacitor in series LCR circuit where L = 1H, C = 20 μF and R = 300Ω and the applied emf has the form $V = 50\sqrt{2} . \sin(100t)$ volts
Options:
(a) 70
(b) 30
(c) 20
(d) 50
Answer: (d)

Question: A graph is plotted between voltage and frequency of photons. Metal 1 and metal 2 are the photosensitive plates.
Statement-I: The slope of the graph is h/e where h is planck's constant and e is electronic charge

Statement-II: Maximum kinetic energy of photoelectron ejected will be more in metal 2 than maximum kinetic energy of photoelectron ejected in metal 1.



Options:
(a) Statement-I is true statement-II is false
(b) Both Statements are correct and statement-II is the correct explanation of S1
(c) Statement-I is false ; statement-II is true
(d) Both statements are false
Answer: (a)

Question: Three capacitors with capacitance $C_1 = 25\mu F$, $C_2 = 30\mu F$ and $C_3 = 45\mu F$ are used as a combination. If C_1, C_2 and C_3 are arranged in parallel combination, total energy stored is E. If C_1, C_2, C_3 are arranged in series combination, total energy stored is E'. Then E/E' is
Options:
(a) 7.5
(b) 8.5
(c) 9.5
(d) 10.5
Answer: (c)

Question: The ratio of electrostatic force to Gravitational force between a proton and an electron is in order
Options:
(a) 10^{28}
(b) 10^{19}
(c) 10^{22}
(d) 10^{8}
Answer: (a)

Question: A block of mass 5kg is kept on the ground. A solid cylinder of 25 kg is kept on the block, after some ground yields and system falls with an acceleration of $0.1 m/s^2$. The reaction force by ground is
($g = 9.8\ m/s^2$)
Options:
(a) 291 N
(b) 294 N
(c) 297 N
(d) 576 N
Answer: (a)

Question: Ratio of radius of gyration of solid cylinder and hollow sphere is
$\frac{K_{sylinder}}{K_{sphere}} = \sqrt{\frac{3}{x}}$
Options:

(a) 4
(b) 5
(c) 6
(d) 7
Answer: (a)

Question: Match the column

1) Escape velocity	A) $\sqrt{gR}$
2) Orbital velocity	B) $\sqrt{2gR}$
3) Gravitational Potential Energy	C) $-\frac{GM_1M_2}{R}$

Options:
(a) 1 - A, 2 - B, 3 - C
(b) 1 - B, 2 - A, 3 - C
(c) 1 - C, 2 - B, 3 - A
(d) 1 - C, 2 - A, 3 - B
Answer: (b)

Question: Find mutual inductance if radius of inner loop is a and outer loop is b
Options:
(a) $\mu_o \pi a^2/2b$
(b) $\mu_o \pi a^2/2b^2$
(c) $\mu_o \pi a/2b$
(d) $\mu_o \pi/2b$
Answer: (a)

Question: A wire of resistance 1Ω, area of cross section $a = 2 \times 10^{-6}$ m^2, resistivity $\rho = 4 \times 10^{-7}$ Ω - m. The wire is horizontal and floats in equilibrium in the presence of a magnetic field B. Wire carries current of 2 Amperes and is of mass 0.5 kg. If B is $x \times 10^{-1}$ T, then find x
Answer: (5)

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[MORNING SHIFT]

Chemistry

Question: Arrange them in decreasing order of boiling point:

(a) $CH_3 - CH_2 - O - CH_2 - CH_2 - CH_3$

(b) $CH_3 - CH_2 - CH_2 - CH_2 - CH_3$

(c) $CH_3 - C - CH_2 - CH_2 - CH_3$

$$\begin{array}{l} CH_3 - \underset{\underset{\displaystyle o}{\|}}{C} - CH_2 - CH_2 - CH_3 \end{array}$$

(d) $CH_3 - C - CH_2 - CH_3 - CH_3$

$$\begin{array}{l} CH_3 - \underset{\underset{\displaystyle OH}{|}}{C} - CH_2 - CH_3 - CH_3 \end{array}$$

Options:
(a) D > C > A > B
(b) D > A > C > B
(c) B > C > A > D
(d) C > A > B > D
Answer: (a)

Question: Ninhydrin test is used for
Options:
(a) Starch
(b) Cellulose
(c) PVC
(d) Egg Albumin
Answer: (d)

Question: Ethylene → no of sigma & pi bonds
Options:
(a) 4-sigma , 1 pi
(b) 5 sigma, 1 pi
(c) 6 sigma, 0 pi
(d) 4 sigma, 2 pi
Answer: (b)

Question: Which is least paramagnetic ?

Options:

(a) $[Fe(H_2O)_6]^{2+}$
(b) $[Cr(H_2O)_6]^{2+}$
(c) $[Mn(H_2O)_6]^{2}$
(d) $[V(H_2O)_6]^{2+}$

Answer: (d)

Question: "In a typical cell used in wall clocks, the following reaction occurs at cathode ".

Options:

(a) Oxidation of Mn from +7 to +2
(b) Oxidation of Mn from +4 to +3
(c) Reduction of Mn from +3 to +4
(d) Reduction of Mn from +2 to +7

Answer: (c)

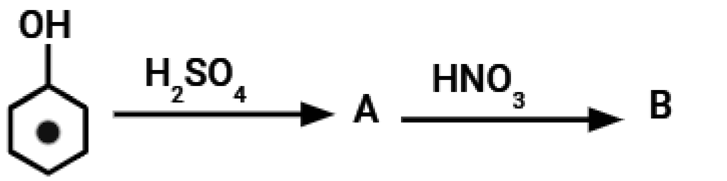
Question: Which has highest no of O.S. ?

Options:

(a) Mn
(b) Co
(c) Ti
(d) Sc

Answer: (a)

Question: Sum of oxygen atoms in A & B.



Options:

(a) 12
(b) 14
(c) 11
(d) 13

Answer: (d)

Question:

Kr	**K**	**$2A + 3B \rightarrow$ Products**
[A]	**[B]**	**Rate**
1.0	**2.4**	**1.6×10^{-2}**
2	**2.4**	**32×10^{-3}**
3	**7**	**8×10^{-4}**
3	**14**	**32×10^{-4}**

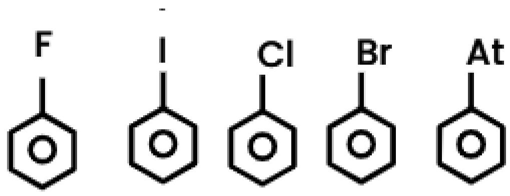
The overall order of reaction is

Options:

(a) 2

(b) 3
(c) 1
(d) 0
Answer: (b)

Question: Which of following(s) will be formed with sandmeyer reaction ?



No of reacting species =
Options:
(a) 3
(b) 4
(c) 2
(d) 1
Answer: (c)

Question: Arrange the following ligands according to their strength:
NH_3 , F^-, Br^-, H_2O
Options:
(a) $H_2O > NH_3 > F^{\ominus} > Br^{\ominus}$
(b) $F^{\ominus} > Br^{\ominus} > NH_3 > H_2O$
(c) $NH_3 > H_2O > F^{\ominus} > Br^{\ominus}$
(d) $Br^{\ominus} > F^{\ominus} > H_2O > NH_3$
Answer: (c)

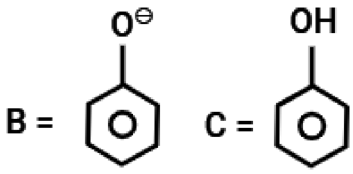
Question: Assertion : Cis alkene is more polar than trans alkene
Reason: Trans But-2-ene is having zero dipole moment.
Options:
(a) Assertion is correct, reason is correct. Reason is the correct explanation of assertion.
(b) Assertion is correct, reason is correct. Reason is the incorrect explanation of assertion.
(c) Assertion and Reason both are incorrect.
(d) Assertion is correct, reason is incorrect. Reason is the correct explanation of assertion.
Answer: (a)

Question: Consider the reaction :
$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2\ Fe(s) + 3CO_2(g)$
Which of the following will not affect the equilibrium state
(i) Addition of Fe_2O_3
(ii) Addition of CO_2
(iii) Decreasing mass of Fe_2O_3
(iv) Removal of CO
Options:
(a) (II) and (IV)
(b) (I) and (IV)

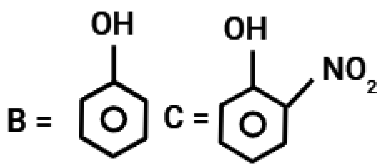
(c) (I) and (III)
(d) All will affect the equilibrium
Answer: (c)

Question: Identify B and C

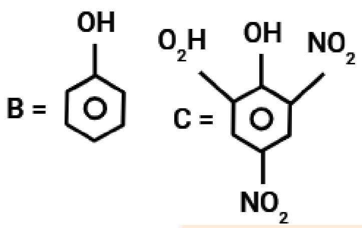
Options:
(a)



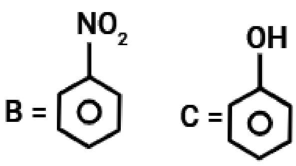
(b)



(c)



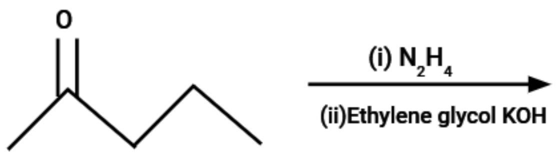
(d)



Answer: (c)

Question: The heat of combustion of solid benzene acid at constant volume is -321.30 KJ at 27°C . The heat of combustion at constant prosecute is (-321.30 -x) kg the value of x is
Options:
(a) 1.57
(b) 1.39
(c) 1.24
(d) 1.45
Answer: (c)

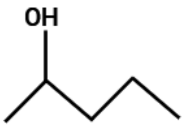
Question:



Options:

(a)

(b)



(c)

(d)

Answer: (a)

Question: Assertion: For 13th group the +1 oxidation state increases down the group.
Reason: The atomic size of Ga is greater than that of Al.

Options:
(a) Assertion-Correct; Reason-Incorrect
(b) Assertion-Correct; Reason-Correct
(c) Assertion-Incorrect ; Reason-Correct
(d) Assertion-Incorrect; Reason-Incorrect
Answer: (a)

Question: Find out the work done in the following process
Options:
(a) 61.5 Joule
(b) -61.5 Joule
(c) +246 Joule
(d) -246 Joule
Answer: (a)

Question: If the number of neutrons in the most abundant isotope of boron is 'x' and its highest oxidation state in unsaturated compound is 'y' then find the value of (x + y)
Options:
(a) 6
(b) 4
(c) 3
(d) 9
Answer: (d)

Question: Which of the following are correct statements for given species O^{2-} , F^- , Na^+ , Mg^{2+}

(a) O^{2-} is largest in size

(b) Mg^{2+} is smallest in size

(c) All have same effective nuclear charge

(d) All are isoelectronic

Options:

(a) a, b and c

(b) a, b and d

(c) b, c and d

(d) a, c and d

Answer: (b)

Question: Arrange the following carbocations in increasing order of their stability

(I) $(CH_3)_2CH^+$

(II) $C\,H_3^+$

(III) $CH_3 - CH_3^+$

(IV) $(CH_3)_3C^+$

Options:

(a) II > III > I > IV

(b) IV > I > III > II

(c) IV > III > I > II

(d) III > I > II > IV

Answer: (b)

Question: Which of the following cation will give green colour in Borax bead test

Options:

(a) Iron

(b) Cobalt

(c) Manganese

(d) Nickel

Answer: (a)

Question: The molar conductivities of a divalent cation (M^{2+}) and monovalent anion (A^-) are 57 S cm^{-1} mol^{-1} and 73 S cm^{-1} mol^{-1} respectively. Then find the total molar conductivity shown by their compound in S cm^{-1} mol^{-1}

Options:

(a) 210 S cm^{-1} mol^{-1}

(b) 203 S cm^{-1} mol^{-1}

(c) 209 S cm^{-1} mol^{-1}

(d) 207 S cm^{-1} mol^{-1}

Answer: (b)

Question: Number of electrons around Nitrogen in lewis dot structure of NO^{2-} (integer)

Answer: 8

Vedantu

JEE-Main-05-04-2024 (Memory Based) [MORNING SHIFT]

Maths

Question: $\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y}$
Options:
(a) π^3
(b) π^2
(c) π
(d) π^4
Answer: (b)

Question: $y(1) = 1$ find $\lim_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{(t-x)}$
Options:
(a) 8
(b) 16
(c) 24
(d) 32
Answer: (c)

Question: If the function $f(x) = \frac{\sin 3x + \alpha \sin x - 3\beta \cos 3x}{x^3}, x \in R,$ is continuous at x = 0, then f(0) is
Options:
(a) 2
(b) -2
(c) 4
(d) -4
Answer: (d)

Question:
$m = \frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} \cdots\cdots \frac{1}{\sqrt{99}} + \frac{1}{\sqrt{100}}$
$n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} \cdots\cdots \frac{1}{99.100}$
Find 2m + n
Options:
(a) 16.99
(b) 17.99

(c) 19.99
(d) 18.99
Answer: (d)

Question: If the length of focal chord of $y^2 = 12x$ is "l"and if the distance of the focal chord from origin is d then ld^2
Options:
(a) 102
(b) 104
(c) 106
(d) 108
Answer: (d)

Question: $Ax^2 + bx + c$, taking the value of a, b, c from the set {1, 2, 3, 4, 5, 6, 7, 8} so that it has repeated roots find probability of it.
Options:
(a) 1/64
(b) 1/32
(c) 1/16
(d) 1/8
Answer: (a)

Question: No. of ways by which sum of the number become 16 by throwing 4 dice
Options:
(a) 120
(b) 125
(c) 130
(d) 135
Answer: (b)

Question: ABCD is a rectangle of sides of length 4 and 2 which is inscribed in rectangle PQRS of area of PQRS is min find $(a+b)^2$ where a and b are sides of rectangle pqrs
Options:
(a) 36
(b) 72
(c) 108
(d) 240
Answer: (b)

Question: $(1+2x-3x^3)\left(\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9\right.$ Constant term in the expansion of
Options:
(a) 1
(b) 1/4
(c) 1/2
(d) 3/4
Answer: (c)

Question: $y=x^2-5x$
$y=7x-x^2$

Find area bounded by the curve
Options:
(a) 36
(b) 72
(c) 108
(d) 240
Answer: (b)

Question: If $\frac{dy}{dx}+2y=\sin\,2x$ and $y\left(0\right)=\frac{3}{4}$, then $y\left(\frac{\pi}{8}\right)$ is equal to
Options:
(a) $e^{\frac{\pi}{8}}$
(b) $e^{\frac{\pi}{6}}$
(c) $e^{\frac{-\pi}{4}}$
(d) None
Answer: (c)

Question: $f(x)=x^5+2x^3+3x+1$
g(x) such that g(f(x))=x for all x
Find g(7)/g'(7)
Options:
(a) 1/4
(b) 1/7
(c) 1/14
(d) 1/8
Answer: (c)

Question: Suppose $\theta\in\left[0,\ \frac{\pi}{4}\right]$ is a solution at 4cosθ - 3sinθ = |then cosθ is equal to
Options:
(a) $\frac{6-\sqrt{6}}{\left(3\sqrt{6}-2\right)}$
(b) $\frac{4}{(3\sqrt{6}+2)}$
(c) $\frac{4}{(3\sqrt{6}-2)}$
(d) $\frac{6-\sqrt{6}}{(3\sqrt{6}+2)}$
Answer: (c)

Question: Lines parallel to coordinate axes passing through (3,2) are tangents to a unit circle which is closer to origin. Find the shortest distance between circle and (5,5)
Answer: 4

Question: $Sinx+3x-\frac{2}{\pi}(x^2+x)$ Check if f is increasing and f' is decreasing $\left(0,\ \frac{\pi}{2}\right)$
Answer: f(x) is increasing and f'(x) is decreasing